

LEARNING OUTCOMES

At the end of this topic you should be able to:

- describe the different types of solutions
- give examples of different types of solutions
- explain the difference between solutions, suspensions and colloids in terms of particle size, sedimentation and light scattering.

DID YOU KNOW?

A *heterogeneous* mixture is one in which the composition is not uniform (the same) throughout. Examples of heterogeneous mixtures are suspensions, e.g. chalk dust in water, and colloids, e.g. mist or fog.

DID YOU KNOW?

Colloids are heterogeneous mixtures because the particles are not always distributed evenly throughout the mixture. Mayonnaise is a heterogeneous mixture of vinegar and oil but it usually has egg yolk added to it to make sure that the colloid does not separate back into separate layers of vinegar and oil.

Solutes, solvents and solutions

When you shake sugar in water, the sugar disappears to form a **solution**. The sugar dissolves in the water. All parts of the solution formed are exactly the same. This is an example of a homogeneous mixture. 'Homogeneous' means the same throughout. The main component of the solution is called the **solvent** and the minor component is called the **solute**.

A solution is a uniform mixture (homogeneous mixture) of two or more substances.

A solute is a substance that is dissolved in a solvent. The solute can be a solid, a liquid or a gas.

A solvent is a substance that dissolves a solute. The solvent can also be a solid, liquid or gas.

Solutions

In solutions:

- The particles are so small that they cannot be seen. They are less than 1 nanometre in diameter. (1 nanometre = 1 millionth of a millimetre.)
- They are generally transparent. Light can be transmitted through them. The exception is a solution of a solid in another solid.
- The particles cannot be separated by filtration, and do not separate out by letting the mixture stand for a while.

Table 2.3.1 Types of solution

Solute	Solvent	Example
Solid	Liquid	Seawater (salt dissolved in water)
Solid	Solid	Bronze (tin dissolved in copper)
Gas	Liquid	Carbonated water (carbon dioxide dissolved in water)
Liquid	Liquid	Rum (ethanol dissolved in a mixture of liquids)
Gas	Gas	Air (mixture of nitrogen, oxygen and other gases)

Suspensions

When you shake powdered clay with water, the mixture appears milky. The clay particles can just about be seen. We call this a **suspension** of clay in water. When you leave the mixture of clay in water for a little while, the clay particles settle to the bottom of the container. The settling of the solid is called **sedimentation**.

A suspension is a mixture of small particles dispersed in another substance, and in which the small particles settle on standing.

In suspensions:

- The particles are large enough to be just visible. They are more than 1000 nanometres in diameter.
- They are opaque (not transparent) to light. Light is scattered by them (see Figure 2.3.1).
- The particles can be separated by filtration. The particles undergo sedimentation – they eventually settle to the bottom of the container in which they are placed.

Colloids

Colloids have some of the properties of solutions but are more like suspensions in many ways. They consist of one type of particle (solid, liquid or gas) dispersed throughout another. An example is mayonnaise, where particles of vinegar are dispersed in vegetable oil (see Figure 2.3.2).

In colloids:

- The particles are not large enough to be visible. They are larger than those in a solution but smaller than those in a suspension, so are between 1 and 1000 nanometres in diameter.
- They are opaque (not transparent) to light. Light is scattered by them (see Figure 2.3.1).
- The particles are too small to be separated by filtration.
- The particles do not undergo sedimentation. They do not settle on standing.

Table 2.3.2 Examples of colloids

Nature of colloid	Type of colloid	Example
Solid dispersed in gas	Aerosol	Smoke particles in air
Gas dispersed in liquid	Foam	Whipped cream
Liquid dispersed in liquid	Emulsion	Milk (fat in water)
Solid dispersed in liquid	Sol	Paint
Liquid trapped in solid	Gel	Jelly

KEY POINTS

- 1 A solute dissolves in a solvent to form a solution.
- 2 A solution has particles that cannot be seen. They are generally transparent and the components cannot be separated by sedimentation.
- 3 A suspension has particles that can be seen. Suspensions scatter light and the components can be separated by sedimentation.
- 4 A colloid has particles intermediate in size between those in a solution and those forming the dispersed part of a suspension. Colloids scatter light and the components cannot be separated by sedimentation.

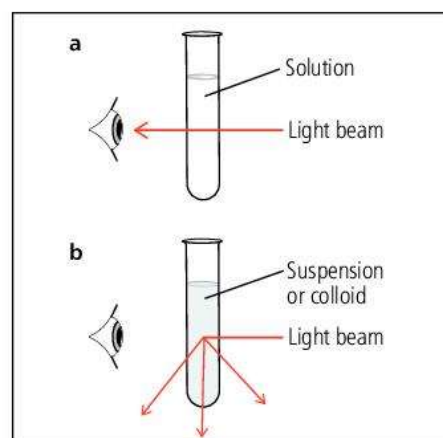


Figure 2.3.1 **a** Light is transmitted through a solution. **b** Light is scattered by the small particles present in a suspension or a colloid. The light is not transmitted directly to the observer.

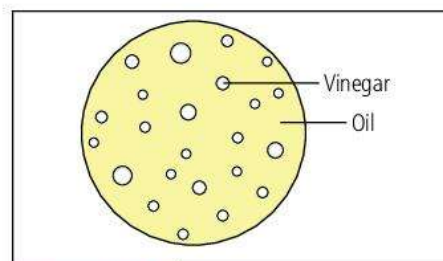


Figure 2.3.2 Mayonnaise is a type of colloid called an emulsion (see Table 2.3.2). Droplets of vinegar are spread throughout the vegetable oil.



Figure 2.3.3 Vinegar is a solution, so is transparent to light. Milk is a colloid, so scatters light and appears opaque.